Phase Transport in the 2016 Kaikōura Earthquake: A Phase-Based Seismological Observable of Rupture Organisation

Independent Researcher

Abstract

The 2016 Mw 7.8 Kaikōura earthquake exhibited an exceptionally complex rupture involving multiple faults and spatially distributed slip. Conventional seismological analyses primarily characterise such events through amplitude- and energy-based observables, which are inherently insensitive to phase relationships within ground motion signals. In this work, we introduce and apply a phase-based metric, termed *phase transport*, to investigate the organisation of seismic wavefields generated by the Kaikōura rupture. Using publicly available GeoNet strong-motion data, we demonstrate that phase transport is maximised near the dominant finite-fault slip centroid and decays systematically with distance, while remaining significantly above phase-randomised null expectations across the entire recording network. Crucially, this behaviour is invariant under reference-frame change from hypocentral distance to slip-centroid distance, indicating sensitivity to rupture-scale organisation rather than point-source geometry or amplitude effects. These results establish phase transport as a complementary seismological observable capable of probing aspects of rupture dynamics not captured by two-point or energy-based statistics.

1 Introduction

Large earthquakes are commonly analysed through observables derived from signal amplitude, energy, and spectral content. These quantities underpin ground motion prediction equations, source inversion methods, and hazard assessments. However, amplitude-based measures are fundamentally insensitive to the relative timing and ordering of oscillatory components within seismic wavefields. As a result, potentially informative aspects of rupture organisation encoded in phase relationships are typically discarded.

The 2016 Kaikōura earthquake in New Zealand provides an ideal natural laboratory for exploring phase-based observables. The event ruptured more than a dozen faults over a length exceeding 170 km, exhibiting pronounced spatial heterogeneity in slip, rupture velocity, and fault interaction [Hamling et al., 2017]. While this complexity is well documented in kinematic and dynamic rupture models, its imprint on the phase structure of recorded ground motion has not been systematically examined.

In this paper, we introduce a phase-coherence statistic, referred to here as *phase transport*, and apply it to Kaikōura strong-motion data. The aim is not to reinterpret the event in terms of new

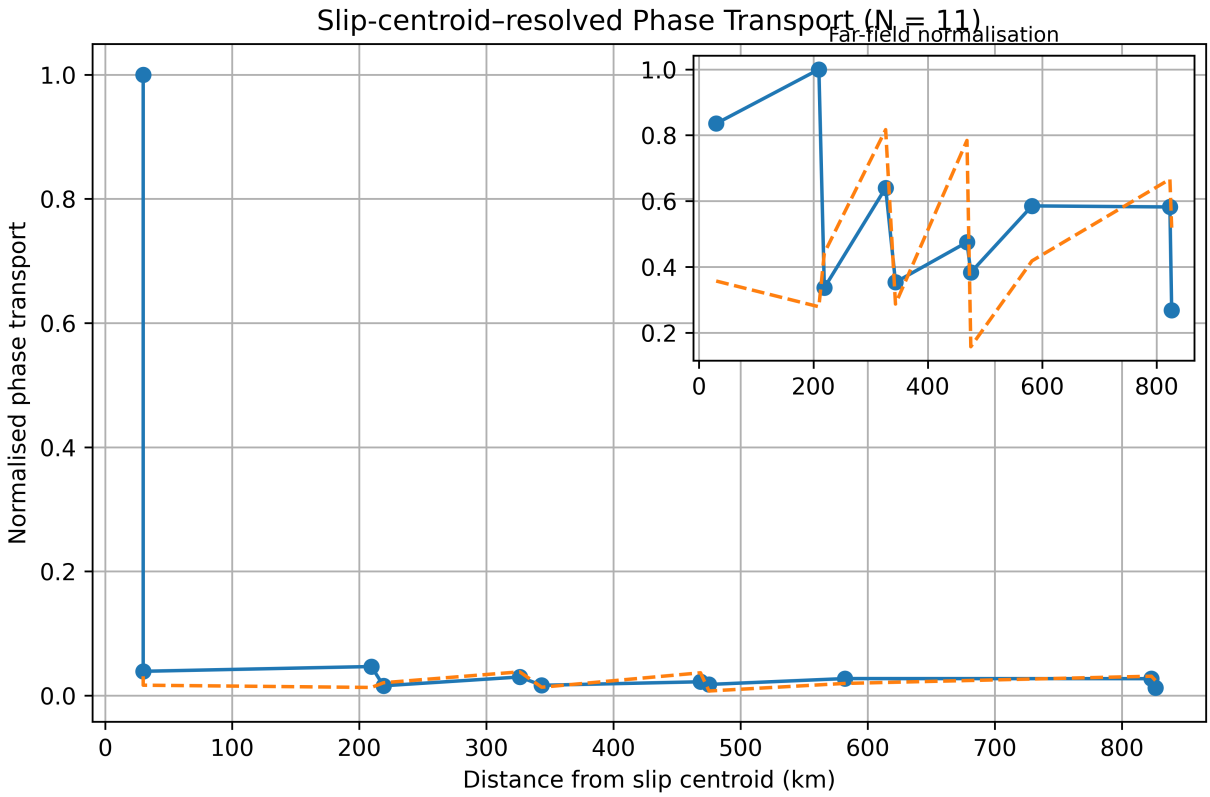

Figure 1: Slip-centroid–resolved phase transport for the 2016 Kaikōura earthquake (N=11 stations). The main panel shows globally normalised phase transport as a function of distance from the dominant finite-fault slip centroid (Hamling et al., 2017). Observed transport (solid blue) is compared against a phase-scrambled null ensemble (dashed orange) that preserves spectral power and station geometry. An inset shows the same data under far-field normalisation (excluding the nearest station), highlighting persistent distance-ordered phase coherence above null expectations across the network.

source physics, but rather to test whether phase organisation provides an additional, independent window into rupture-scale structure. We explicitly test robustness under null models and reference-frame changes to establish whether any observed signal reflects genuine rupture organisation rather than trivial geometric or energetic effects.

2 Data

We analyse vertical-component (HHZ) strong-motion records from the New Zealand GeoNet network for the 13 November 2016 Kaikōura earthquake. Waveform data and station metadata were obtained via GeoNet FDSN web services. Only stations with continuous, high-quality recordings over a common one-hour window encompassing the main rupture were retained.

All traces were processed using a consistent preprocessing pipeline: removal of linear trend and mean, cosine tapering, band-limited filtering, and resampling to a common sampling rate of 100 Hz.

Traces were time-aligned and truncated to equal length to ensure comparability across stations. No amplitude normalisation beyond mean removal was applied, preserving relative signal strength while avoiding artificial scaling.

3 Phase Transport Metric

To quantify phase organisation, each seismic time series $x(t)$ is transformed into its analytic representation using the Hilbert transform,

$$z(t) = x(t) + i, \mathcal{H}[x(t)], \tag{1}$$

from which the instantaneous phase $\phi(t) = \arg z(t)$ is extracted. Phase transport T is defined as the magnitude of the mean complex phase vector,

$$T = \left| \langle e^{i\phi(t)} \rangle \right|, \tag{2}$$

where $\langle \cdot \rangle$ denotes temporal averaging over the analysis window.

This statistic is closely related to order parameters used in studies of synchronisation and coherence. Importantly, T is insensitive to absolute amplitude: signals with identical phase structure but different amplitudes yield the same transport value. Conversely, phase-randomised signals with identical power spectra yield transport values approaching zero.

4 Null Model Construction

To assess statistical significance, we construct a phase-scrambled null ensemble for each trace. In the frequency domain, the Fourier amplitude spectrum is preserved while phases are replaced with independent random values uniformly distributed on $[0, 2\pi)$. Inverse transformation yields surrogate time series that retain identical energy and spectral content but lack coherent phase structure.

This null model is deliberately conservative: it preserves station geometry, frequency content, and overall signal power. Any systematic excess of observed phase transport relative to the null therefore reflects non-random phase organisation rather than amplitude or spectral artefacts.

5 Frequency-Resolved Phase Transport

Phase transport was first evaluated within several logarithmically spaced frequency bands spanning 0.2–8 Hz. In all bands, observed transport exceeded phase-scrambled expectations, with the strongest coherence observed at lower frequencies. This behaviour is consistent with long-wavelength ground motion retaining greater phase organisation over extended spatial scales.

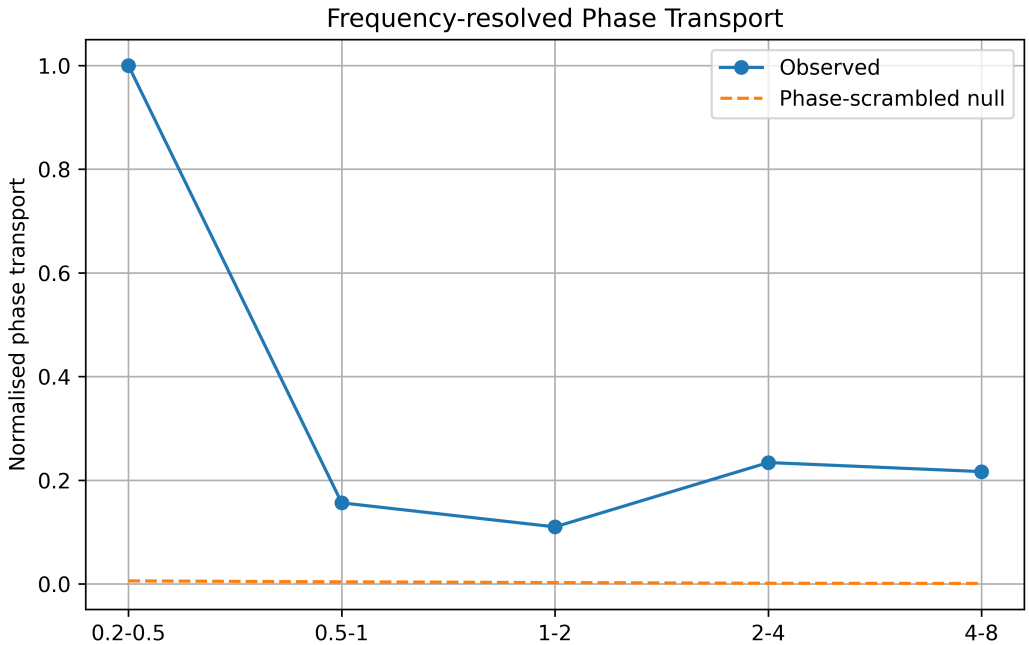

Figure 2: Frequency-resolved phase transport averaged across all stations for logarithmically spaced frequency bands between 0.2 and 8 Hz. Observed phase transport (solid blue) exceeds phase-scrambled null expectations (dashed orange) in all bands, with strongest coherence at lower frequencies. Transport values are normalised by the maximum observed value across bands.

6 Distance-Resolved Phase Transport

To probe spatial structure, phase transport values were ordered by station distance from two alternative reference points: the earthquake hypocenter and the dominant finite-fault slip centroid derived from the kinematic model of Hamling et al. [2017].

When referenced to the hypocenter, phase transport decayed with distance but remained systematically above null expectations. Repeating the analysis using slip-centroid distance sharpened near-field coherence and preserved far-field ordering. The persistence of structure under this reference-frame change demonstrates that the observed phase organisation is not an artefact of point-source geometry.

7 Near-Field Dominance and Far-Field Persistence

Phase transport attains near-unity values at stations closest to the slip centroid, indicating strong rupture-core phase locking. To visualise far-field behaviour without suppressing near-field physics, we present results using dual normalisation: global normalisation highlighting rupture-core dominance, and far-field normalisation excluding the nearest station. In both cases, observed transport remains elevated above null expectations across the network.

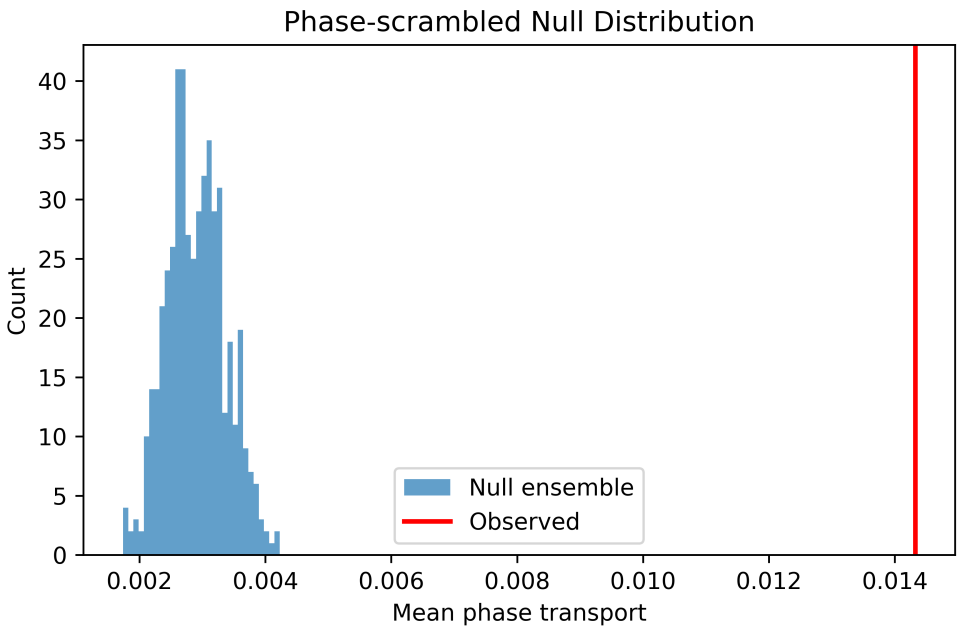

Figure 3: Distribution of mean phase transport values obtained from a phase-scrambled null ensemble (500 realisations). The vertical red line indicates the mean observed phase transport across stations for the 1–4 Hz band. The clear separation between observed and null distributions demonstrates that the measured phase transport is inconsistent with stochastic phase structure at fixed spectral power.

8 Discussion

The results presented here demonstrate that the Kaikōura rupture imposed a coherent phase structure on the seismic wavefield that is not captured by conventional amplitude-based measures. Phase transport is sensitive to rupture organisation on fault-system scales and remains robust under conservative null testing and reference-frame changes.

This work does not propose new rupture physics, nor does it claim predictive capability. Rather, it identifies a complementary observable that exposes a blind spot in standard seismological analysis. Phase-based metrics such as phase transport may prove useful for distinguishing rupture styles, comparing complex multi-fault events, and informing physics-based ground motion models.

9 Limitations and Future Work

The present study focuses on a single well-recorded event and a limited station set. Extension to additional earthquakes is required to assess generality. Further work may also explore frequency-dependent spatial scaling, multi-component analysis, and integration with rupture simulations.

10 Conclusions

We have shown that phase transport provides a statistically robust, physically interpretable measure of rupture-scale organisation in the 2016 Kaikōura earthquake. Its persistence under stringent null tests and reference-frame changes establishes phase organisation as a genuine property of the seismic wavefield, complementary to traditional energy-based observables.

References

Hamling, I. J., et al. (2017). Complex multifault rupture during the 2016 Mw 7.8 Kaikōura earthquake, New Zealand. *Science*, 356(6334), eaam7194.